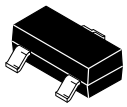
High-Voltage General-Purpose Diode

MMBD1404A



SOT-23
CASE 318-08

Description

Sourced from process 2 V.

ABSOLUTE MAXIMUM RATINGS

(T_A = 25 °C unless otherwise noted) (Notes 1, 2)

Rating	Symbol	Value	Unit
Working Inverse Voltage	W_{IV}	175	V
Average Rectified Current	I_O	200	mA
DC Forward Current	I_F	600	mA
Recurrent Peak Forward Current	i_f	700	mA
Non-Repetitive Peak Forward Surge Current Pulse Width = 1.0 second Pulse Width = 1.0 microsecond	$i_{f(surge)}$	 1.0 2.0	A
Storage Temperature Range	T_{STG}	−55 to +150	°C
Operating Junction Temperature	T_J	150	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. These ratings are based on a maximum junction temperature of 150 °C.
2. These are steady-state limits. **onsemi** should be consulted on applications involving pulsed or low-duty-cycle operations.

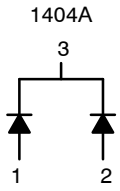
THERMAL CHARACTERISTICS

(T_A = 25 °C unless otherwise noted) (Note 3)

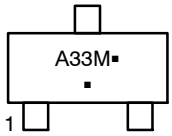
Characteristic	Symbol	Value	Unit
Power Dissipation	P_D	350	mW
Derate Above 25 °C		2.8	mW/° C
Thermal Resistance, Junction-to-Ambient	$R_{\theta JA}$	357	°C/W

3. Device is mounted on glass epoxy PCB 1.6 inch × 1.6 inch × 0.06 inch, mounting pad for the collector lead minimum 0.93 in^2.

CONNECTION DIAGRAM



MARKING DIAGRAM



A33 = Specific Device Code
M = Date Code
▪ = Pb-Free Package

(Note: Microdot may be in either location)

ORDERING INFORMATION

See detailed ordering and shipping information on page 2 of this data sheet.

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ORDERING INFORMATION

Device	Marking	Package	Shipping†
MMBD1404A MMBD1404A-NC	A33	SOT-23 (Pb-Free)	3,000 / Tape & Reel

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

* S, V and NSV Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable.

ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)

Parameter	Symbol	Condition	Min	Max	Unit
Breakdown Voltage	B_V	I_R = 100 μA	250	–	V
Reverse Current	I_R	V_R = 120 V	–	40	nA
		V_R = 175 V	–	100	
Forward Voltage	V_F	I_F = 10 mA	–	800	mV
		I_F = 50 mA	760	920	mV
		I_F = 200 mA	–	1.1	V
		I_F = 300 mA	–	1.25	V
Diode Capacitance	C_O	V_R = 0, f = 1.0 MHz	–	2.0	pF
Reverse Recovery Time	t_{rr}	I_F = I_R = 30 mA, I_{RR} = 3.0 mA, R_L = 100 Ω	–	50	ns

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

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TYPICAL CHARACTERISTICS

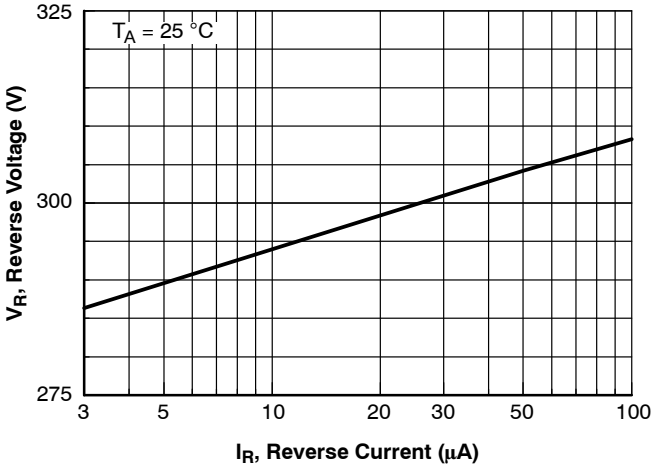


Figure 1. Reverse Voltage vs. Reverse Current
B_V – 1.0 to 100 μA

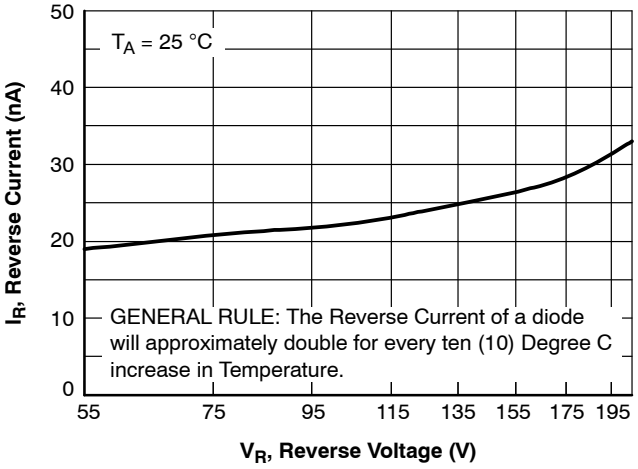


Figure 2. Reverse Current vs. Reverse Voltage
I_R – 55 to 205 V

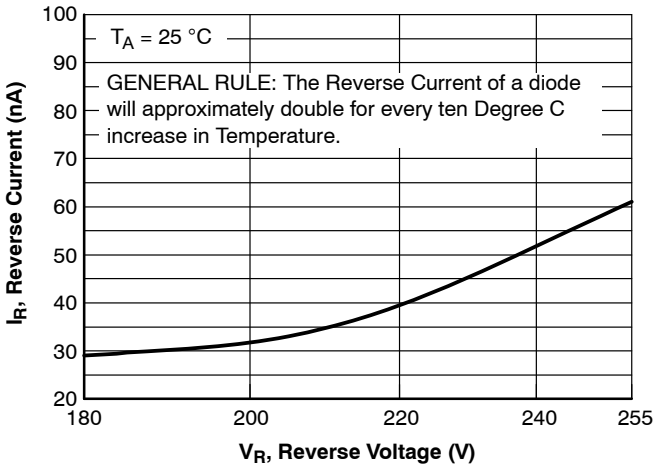


Figure 3. Reverse Current vs. Reverse Voltage
I_R – 180 to 255 V

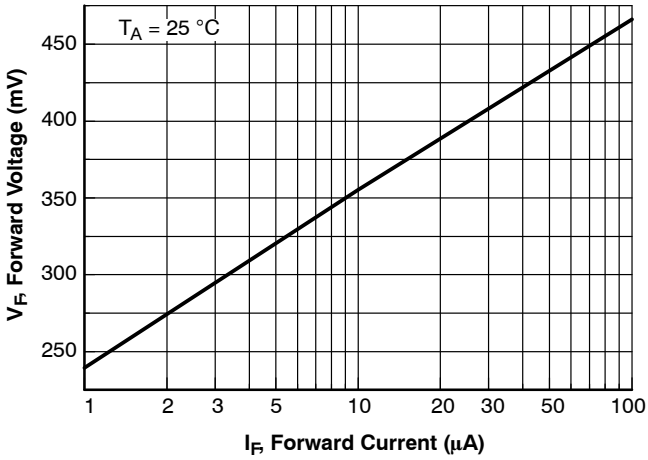


Figure 4. Forward Voltage vs. Forward Current
V_F – 1.0 to 100 μA

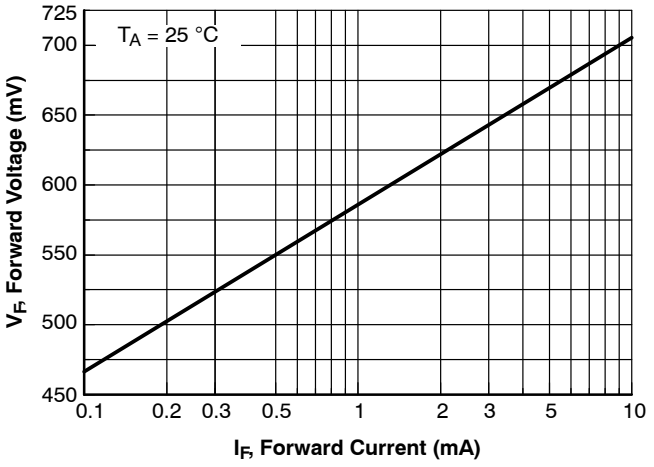


Figure 5. Forward Voltage vs. Forward Current
V_F – 0.1 to 10 mA

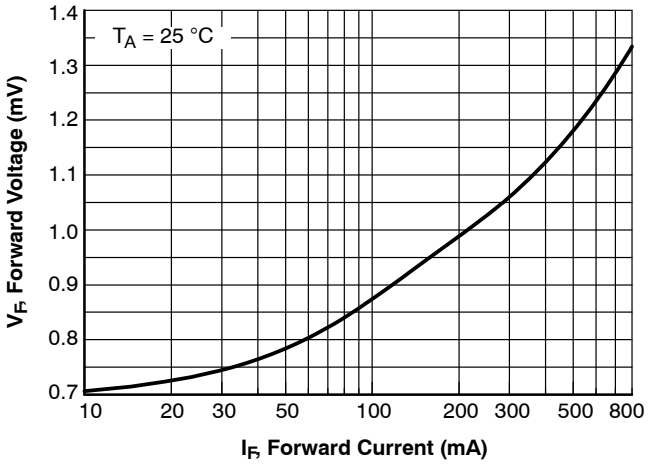


Figure 6. Forward Voltage vs. Forward Current
V_F – 10 to 800 mA

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TYPICAL CHARACTERISTICS (continued)

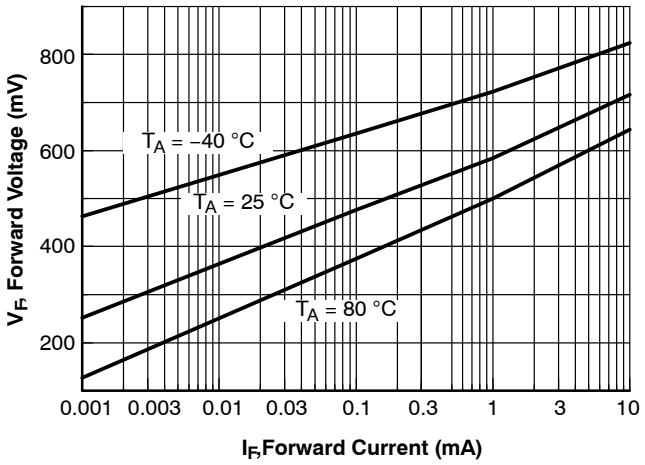


Figure 7. Forward Voltage vs. Ambient Temperature, V_F – 1.0 μA – 10 mA (−40 to +80 °C)

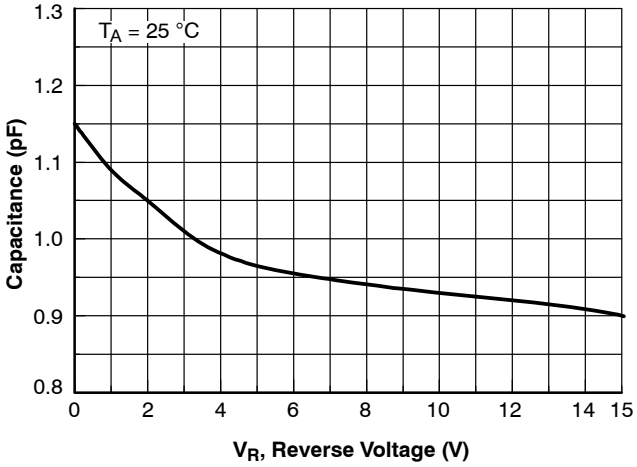


Figure 8. Capacitance vs. Reverse Voltage

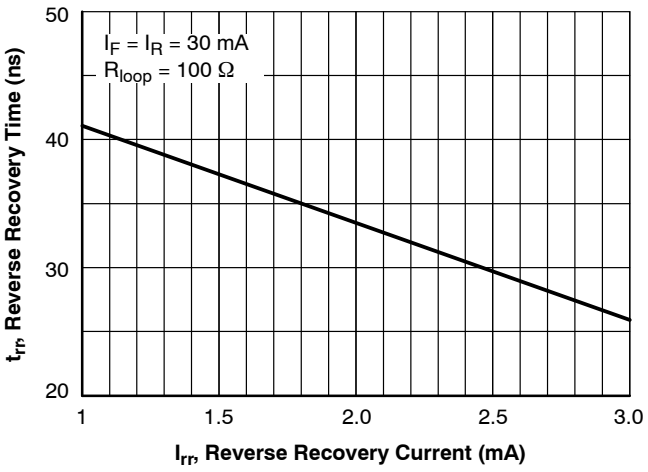


Figure 9. Reverse Recovery Time vs. Reverse Recovery Current (I_{rr})

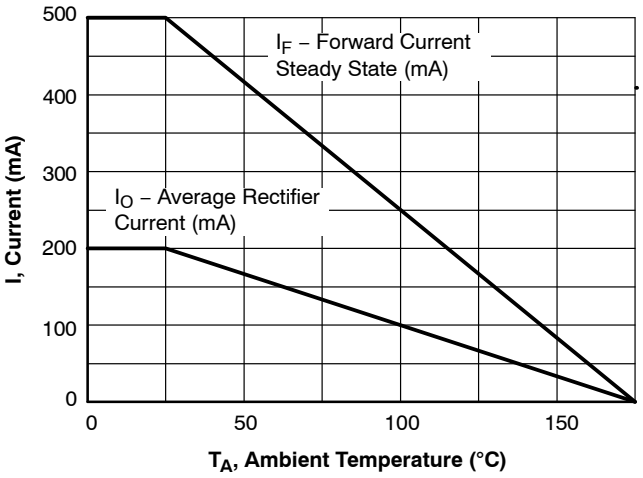


Figure 10. Average Rectified Current (I_O) and Forward Current (I_F) vs. Ambient Temperature (T_A)

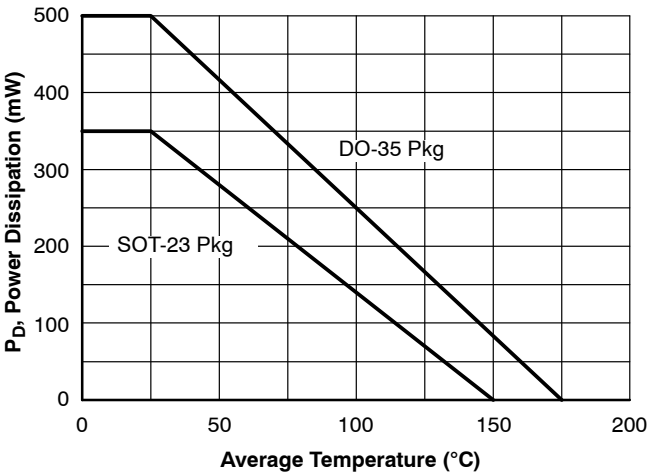


Figure 11. Power Derating Curve

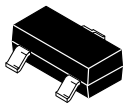
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REVISION HISTORY

Revision	Description of Changes	Date
4	Added “-NC” OPNs to order table.	10/6/2026

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

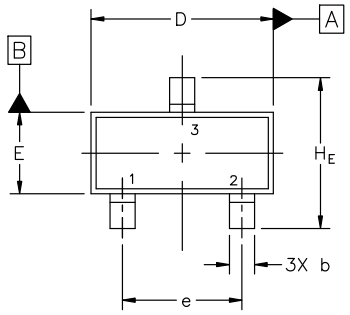
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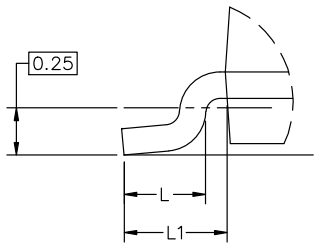
SCALE 4:1

SOT−23 (TO−236) 2.90x1.30x1.00 1.90P
CASE 318
ISSUE AU

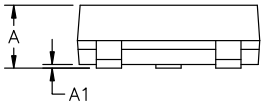
DATE 14 AUG 2024



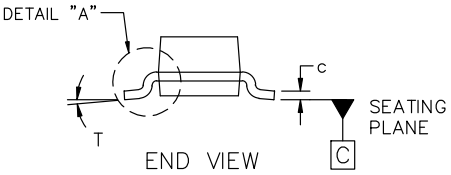
TOP VIEW



DETAIL "A"
Scale 3:1



SIDE VIEW

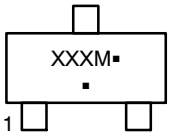


END VIEW

MILLIMETERS			
DIM	MIN	NOM	MAX
A	0.89	1.00	1.11
A1	0.01	0.06	0.10
b	0.37	0.44	0.50
c	0.08	0.14	0.20
D	2.80	2.90	3.04
E	1.20	1.30	1.40
e	1.78	1.90	2.04
L	0.30	0.43	0.55
L1	0.35	0.54	0.69
H_E	2.10	2.40	2.64
T	0°	−−−	10°

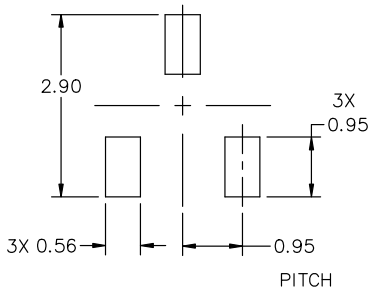
NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSIONS: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.

GENERIC
MARKING DIAGRAM*



XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.



RECOMMENDED
MOUNTING FOOTPRINT

* For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

STYLES ON PAGE 2

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SOT−23 (TO−236) 2.90x1.30x1.00 1.90P
CASE 318
ISSUE AU

DATE 14 AUG 2024

STYLE 1 THRU 5:
CANCELLED

STYLE 6:
PIN 1. BASE
2. EMITTER
3. COLLECTOR

STYLE 7:
PIN 1. EMITTER
2. BASE
3. COLLECTOR

STYLE 8:
PIN 1. ANODE
2. NO CONNECTION
3. CATHODE

STYLE 9:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 10:
PIN 1. DRAIN
2. SOURCE
3. GATE

STYLE 11:
PIN 1. ANODE
2. CATHODE
3. CATHODE−ANODE

STYLE 12:
PIN 1. CATHODE
2. CATHODE
3. ANODE

STYLE 13:
PIN 1. SOURCE
2. DRAIN
3. GATE

STYLE 14:
PIN 1. CATHODE
2. GATE
3. ANODE

STYLE 15:
PIN 1. GATE
2. CATHODE
3. ANODE

STYLE 16:
PIN 1. ANODE
2. CATHODE
3. CATHODE

STYLE 17:
PIN 1. NO CONNECTION
2. ANODE
3. CATHODE

STYLE 18:
PIN 1. NO CONNECTION
2. CATHODE
3. ANODE

STYLE 19:
PIN 1. CATHODE
2. ANODE
3. CATHODE−ANODE

STYLE 20:
PIN 1. CATHODE
2. ANODE
3. GATE

STYLE 21:
PIN 1. GATE
2. SOURCE
3. DRAIN

STYLE 22:
PIN 1. RETURN
2. OUTPUT
3. INPUT

STYLE 23:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 24:
PIN 1. GATE
2. DRAIN
3. SOURCE

STYLE 25:
PIN 1. ANODE
2. CATHODE
3. GATE

STYLE 26:
PIN 1. CATHODE
2. ANODE
3. NO CONNECTION

STYLE 27:
PIN 1. CATHODE
2. CATHODE
3. CATHODE

STYLE 28:
PIN 1. ANODE
2. ANODE
3. ANODE

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